MILITARY SPECIFICATION

ANTENNA GROUP, TACAN, OE-273(V)/URN

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers production of a Tactical Air Navigation (TACAN) all-band shipboard antenna group, OE-273(V)/URN, hereinafter referred to as the antenna group, consisting of an antenna and an antenna control unit for use in conjunction with a TACAN beacon system to provide navigational information to aircraft.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-17/86	Cable, Radio Frequency, Flexible Coaxial, 50 Ohms, M17/86-00001 And M17/86-00002 (Armored)
MIL-S-901	Shock-Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-C-22750	Coating, Epoxy-polyamide
MIL-S-22885	Switches, Push Button, Illuminated, General Specification For
MIL-P-24441	Paint, Epoxy-polyamide, General Specification For

STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, Specifications, Standards And In Technical Documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-003-121), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5985

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MIL-STD-108	Enclosures For Electric And Electronic Equipment; Definition Of And Basic Requirements For
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-291	Standard Tactical Air Navigation (TACAN) Signal
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-794	Parts And Equipment, Procedures For Packaging Of
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
MIL-STD-1387	Procedures For Submission Of Application For Approval Of Nonstandards General Purpose Electronic Test Equipment
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

HANDBOOKS

MIL-HDBK-235	Electromagnetic (Radiated) Environment Considerations For Design And Procurement Of Electrical And Electronic Equipment
MIL-HDBK-241	Design Guide For EMI Reduction In Power Supplies

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

503486	Shipboard TACAN Antenna Group, OE-273(V)/URN And OE-273A(V)/URN
5034290	Shipboard TACAN Antenna Group, OE-273(V)/URN And OE-273A(V)/URN
5034291	Shipboard TACAN Antenna Group, OE-273(V)/URN And OE-273A(V)/URN

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536E	Process Specification-Procedures And Requirements For Preparation And Soldering Of Electrical Connections

NAVAL PERSONNEL COMMAND (NAVPERS)

NAVPERS 18068	Manual Of Navy Enlisted Manpower And Personnel Classifications And Occupational Standards

DAVID W. TAYLOR NAVAL SHIP RESEARCH AND DEVELOPMENT CENTER (DTNSRDC)

Report No. C-3753	Concepts And Interim Procedures For The Design/Analysis Of Exposed Shipboard Equipment To Resist Nuclear Air Blast

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(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

DRAWINGS

INTERNATIONAL TELEPHONE AND TELEGRAPH (AVIONICS) (ITTAV)

1098792	Performance Test Procedure For Antenna Group OE-273(V)/URN (latest revision)
1098793	Performance Test Data For Antenna OE-273(V)/URN (latest revision)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. This equipment shall be in accordance with MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample or samples shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials and processes paragraph of MIL-E-16400, except that prior to Procedure II, selection and request for approval of nonstandards parts and materials, MIL-STD-965, shall be followed. In MIL-STD-275, the minimum spacing between conductors on coated printed wiring boards (PWBs) (applicable to all altitudes) may be 0.381 millimeter (mm) (0.015 inch (in.)) at 80 volts (V) for NAVSEA DWG 503486.

3.2.1 Parts. Parts shall be as specified in 3.2.1.1. through 3.2.1.5.

3.2.1.1 Nonstandard parts. All nonstandard parts shall be approved by the procuring activity prior to first article inspection.

3.2.1.2 Microelectronic devices. All microelectronic devices employed in the design of the antenna group shall conform to MIL-STD-454, Requirement 64.

3.2.1.3 Control locking devices. Control locking devices shall be designed in accordance with the Controls paragraph of MIL-E-16400, except that switches or other controls having detent mechanisms need not have locking devices.

3.2.1.4 Mating connector plugs. Mating connector plugs for all multicontact connector receptacles to be furnished with deliverable equipments shall be Bendix crimp connectors of the 10-2146 series, or equivalent.

3.2.1.5 Telephone jacks. Sound powered telephone circuits and jacks (Type H-27A with cap and retaining chain) shall be provided for communications between the antenna portion of the group and the control unit (see 6.4.4).

3.2.2 Processes. Processes shall be as specified in 3.2.2.1 and 3.2.2.2.

3.2.2.1 Painting. The exterior of the equipment, except for the antenna radome, shall be finished in accordance with the Painting paragraph of MIL-E-16400. The antenna radome shall be painted black with one coat of 0.05 mm to 0.08 mm (0.002-in. to 0.003-in.) thickness of epoxy low gloss No. 3 by Devoe Reynolds Devron 219, or equivalent.

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3.2.2.2 Soldering. Soldering shall be in accordance with WS-6536E.

3.3 Environmental requirements. Environmental requirements shall apply, where applicable, for both the sheltered and unsheltered portions of the antenna group.

3.3.1 Temperature requirements. Temperature Range 2 and Range 4 of MIL-E-16400 shall apply except that the Range 2 high temperature operation shall be +95° Celsius (C).

3.3.2 Salt fog (spray). The salt fog requirements shall be as specified in the Salt fog (spray) paragraph of MIL-E-16400 for both the exposed and sheltered portions.

3.3.3 Nuclear air blast. Conformance to the nuclear air blast requirement shall be achieved by analysis in accordance with methods and procedures specified in DTNSRDC Report C-3753. The level shall be:

Peak overpressure:	3.0 per square inch (psi)
Duration of positive phase:	3.5 seconds
Peak dynamic pressure:	0.2 psi

3.3.4 Lightning protection. A discharge protection rod shall be provided. The rod shall be hard-mounted to the antenna pedestal, positioned at the bow end of the antenna. The peak 135 hertz (Hz) azimuth error attributed to the rod shall not exceed 1.5 degrees in any area, nor will the peak error of the antenna group including all sources exceed 3.0 degrees peak. The peak 15 Hz bearing error of the system including the errors contributed by the lightning arrester shall not exceed 2.0 degrees total, nor shall the peak error including all sources exceed 10 degrees.

3.3.5 Ice loading and heavy rain. Icing and rain build-up shall be minimized by ensuring as smooth a surface as possible.

3.3.6 Humidity. Humidity requirements shall be in accordance with MIL-STD-810, Method 507.2, Procedure II.

3.3.7 Shock. The antenna and control unit shall be capable of withstanding the Grade A, Type A, Class 1 shock test of MIL-S-901.

3.3.8 Vibration. The antenna and control unit shall be capable of withstanding the Type 1 vibration test of MIL-STD-167-1.

3.4 Electrical requirements. Electrical requirements shall be as specified in 3.4.1 through 3.4.1.2.

3.4.1 Power supply. The antenna group shall be capable of operating from a primary power source of 115 V, single phase 60 Hz or 115 V, single phase 400 Hz. No modifications, adjustments, cable changes, and so forth shall be necessary to achieve this capability. The power supply outputs shall be protected by automatic cutouts during overvoltage and overcurrent. The power supplies shall not be damaged and shall return to normal operation upon removal of abnormal conditions. Power supplies shall be operated at no more than 80 percent of their rated values. A switch controlled voltmeter shall be included on the control unit front panel to monitor direct current (DC) power supply output voltages. The antenna group shall comply with the contraints of DoD-STD-1399, Section 300, to avoid introducing anomalies into the power distribution system.

3.4.1.1 Power variations. Satisfactory operation shall be maintained with maximum power line variations as specified in the Power service variations paragraph of MIL-E-16400 and including the characteristics as specified for DoD-STD-1399, Section 300, Type I.

3.4.1.2 Harmonic current limitation. The equipment shall conform to the limits specified in the Harmonic current limitation paragraph of MIL-E-16400.

3.5 Maintenance design. Maintenance design shall be as specified in 3.5.1 through 3.5.2.

3.5.1 Test provisions. Test provisions shall conform to MIL-STD-454, Requirement 32, except as specified in 3.5.1.1 through 3.5.3.

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3.5.1.1 Test point criteria. Except for shop testing, test points shall be accessible as specified in a through c:

a. With the removal of no more than one cover panel
b. While the equipment is operational, and
c. Without removal of any cables or connectors

The use of extender boards is permitted to allow access to test points on printed circuit cards. Test points shall be provided for the requirements specified in d through g. (Multiple use of test points is permitted.)

d. Signal quality and circuit performance. Test points, test jacks, or both, shall be provided to permit the injection of signals and the monitoring of signals at the input/output (I/O) terminals of the system, each unit and each assembly (see 6.4.3).

e. Alignment. Test points, test jacks, or both, shall be provided to energize, inject a standard dynamic signal, or monitor the intermediate circuit affected by the alignment. Where test measurement and diagnostic equipment (TMDE) is used to align or adjust a system, unit, assembly, and subassembly (see 6.4.2), circuitry shall be designed such that a technician can see and operate the TMDE while making adjustments and probing the circuit under test.

f. Fault isolation. Test points, test jacks, or both, shall be provided to permit the injection and measurement of signals at I/O terminals of the unit, assembly and subassembly to determine satisfactory operation or malfunction of the unit, assembly, or subassembly. Antenna parasitic card faults shall be locatable at the control unit by internal switches and indicator lights only.

g. Shop testing. Shop testing of assemblies and subassemblies will be performed with the assemblies and subassemblies removed from the prime equipment. Test points shall be provided so that the assembly can be as specified in 1 through 3:

1. Activated (such as DC and dynamic signals applied),
2. Checked for quality of operation and
3. Fault isolated to the shop replaceable item (that is, flatpack, voltage controlled oscillator, mixer, detector, multiplier and TO-5 can)

Assembly and subassembly test points shall be provided in a single connector. Assembly and subassembly test points, when approved by the procuring activity, may be placed at various nodes or locations on the assembly and subassembly.

3.5.1.2 General purpose electronic test equipment (GPETE). GPETE for use at all levels of maintenance shall be selected from the standard and substitute standard GPETE listed in MIL-STD-1364. The procedures of MIL-STD-1387 shall apply when the GPETE listed in MIL-STD-1364 will not conform to the testing requirements.

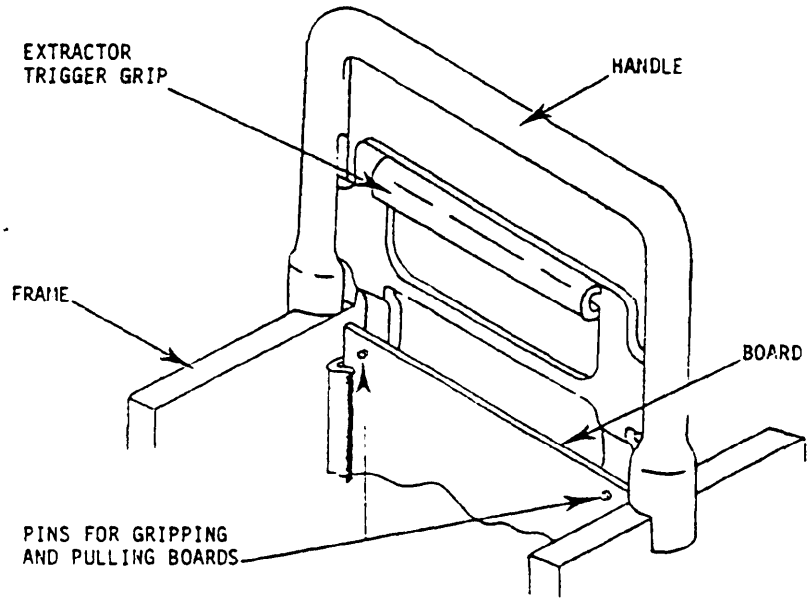
3.5.1.3 Monitor test switch. A monitor test switch shall be provided on the front panel of the control unit. When the switch is engaged, a malfunction shall be introduced into the monitor circuitry to cause all the front panel failure indicators to become illuminated, with the exception of the relative radio frequency (RF) power indicator which shall be activated by adjustment of the RF power alarm threshold selector switch.

3.5.1.4 Test cables. Any extension and test cables or cards required to maintain and troubleshoot this group shall be provided and suitably mounted within the equipment.

3.5.2 Special tools. The need for special tools (see 6.4.6) shall be kept to a minimum. If required, special tools shall be supplied in accordance with MIL-STD-454, Requirement 63 and shall conform to the storage and operating environmental conditions specified for the prime equipment.

3.5.3 Accessibility. Accessibility to the electronics of the mast-mounted portion of the antenna group shall be provided to the greatest extent practicable. Hinged access covers, or equivalent, properly sealed, and incorporating captive quick-release fasteners, shall be provided, to permit troubleshooting and repair of the antenna electronics at the point of installation on the mast. All capture fasteners shall be large enough to be finger-operable. Doors and hinged chassis designed to swing out or down for maintenance access shall have braces and capture locking devices to secure the assembly in the maintenance position. All circuit cards shall be removable by a device similar to that shown in FIGURE 1, and if retaining hardware is used, the device shall be captive and finger-operable. A properly mounted card extender board shall be supplied.

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Recommended extractor tool equivalent to the types commercially available from Flotron Industries and Protolab (Positive Pin-Type pick up with spring loaded trigger).

FIGURE 1. Extractor tool.

3.6 Mechanical design. Mechanical design shall be as specified in 3.6.1 through 3.6.12.

3.6.1 Antenna. The mast-mounted portion of the antenna group shall be configured for top-of-the-mast-mounting.

3.6.2 Control unit. The control unit for the antenna group shall conform to NAVSEA Drawing (DWG) 5034291.

3.6.3 Mast-mounting. The mast-mounted unit shall be provided with a horizontal mounting flange 406.4 mm (16 in.) in overall diameter with eight equally spaced mounting holes, 17.46-mm diameter (0.6875-in. diameter), spaced on a 355.6 mm (14.00 in.) concentric to concentric circle. The eight-hole pattern shall have two opposite holes aligned with the north-south axis of the antenna unit and be drilled to an accuracy of 0.5 degree relative to the north-south axis in accordance with NAVSEA DWG 5034290.

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3.6.4 Radome. The radome shall be constructed such that it has a removable top cover for servicing.

3.6.5 Enclosure. The control unit shall be dripproof 45 degrees and the antenna shall be watertight as specified in MIL-STD-108. NAVSEA DWG 5034290 applies for the pedestal connectors.

3.6.6 Moisture protection. The antenna shall contain a desiccant container and replaceable desiccant capable of controlling moisture within the humidity alarm limits of 3.7.2.1 over a 9-month period under operational conditions. The desiccant shall be accessible from the underside of the antenna. A drain hole shall be provided through the mounting flange of the antenna base and shall be sealed with an allen-head cap screw.

3.6.7 External connections. All externally mounted cables shall be watertight and shall conform to the requirements of the External connection paragraph of MIL-E-16400.

3.6.8 Cable entrances. All cable entrancers shall conform to NAVSEA DWG 5034290 and NAVSEA DWG 5034291.

3.6.9 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators and panel layout paragraph of MIL-E-16400, except that color coding of lamps and indicators shall be in accordance with MIL-STD-1472 and the color coding marking of pin jacks and other test points shall be in accordance with MIL-STD-415. The abbreviations of MIL-STD-12 shall be used for equipment labeling where applicable.

3.6.10 Control illumination. Controls, switches, and dials shall not be illuminated except for those required to change channels, set the magnetic variation, and display LOCAL/REMOTE status. The LOCAL/REMOTE control switch and the BAND switch shall be selected in accordance with MIL-S-22885. The reset switch shall be a nonilluminated pushbutton switch.

3.6.11 Time totalizing meter. A time totalizing meter is required and shall display five significant digits in hours

3.6.12 Fuse locations. All fuses and a spare for each shall be located on front panels. Combined fuseholders and indicators shall be used.

3.7 Specific requirements. Specific requirements shall be in accordance with 3.7.1 through 3.7.3.

3.7.1 Antenna unit. Specific requirements for the antenna unit shall be in accordance with 3.7.1.1 through 3.7.1.3.10.

3.7.1.1 General. The antenna shall include the necessary central array, parasitic cards, switching and monitoring circuitry, monitor probe(s), radome, pedestal, and dessicant. The antenna shall provide a TACAN signal in accordance with MIL-STD-291 (except as modified herein) and properly oriented with respect to true north. Active electronic components shall be kept to a minimum within the antenna itself. As much circuitry as possible shall be accessible from the underside without having to remove the top radome. All circuitry shall be removable without removing the antenna from the mast.

3.7.1.2 Frequency coverage. The antenna shall conform to the electrical performance requirements specified in 3.7.1.3 on all frequencies between (and including) 962 megahertz (MHz) to 1213 MHz. No modifications, adjustments, cable changes, and so forth, shall be made at the antenna to achieve the frequency coverage required. Manual and instantaneous frequency band switching (controllable from the beacon) shall be provided if necessary to conform to all band requirements (see 3.7.2.2).

3.7.1.3 Electrical performance. The electrical performance shall be in accordance with 3.7.1.3.1 through 3.7.1.3.10.

3.7.1.3.1 Polarization. The antenna shall radiate a vertically polarized signal.

3.7.1.3.2 Composite 15-Hz and 135-Hz variable bearing signal. The bearing signal shall be generated by electronically rotating a directional pattern which produces, at a point in space, a composite amplitude modulation of the RF pulse signals at 15 Hz and 135 Hz. As viewed from above, the pattern is rotated clockwise. In the absence of harmonics, the modulation envelope of the detected RF pulse signal shall conform to the following formula.

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$$y = 1.0 + A \sin (2\pi ft \pm \theta - \gamma) + B \sin (18\pi ft \pm \phi - 9\gamma)$$

Where:

a. y = normalized composite 15 Hz to 135 Hz signal amplitude
b. A = modulation index of 15 Hz component
c. B = modulation index of 135 Hz component
d. θ = deviation in azimuth degrees from coincidence of the 15-Hz modulation and the main reference group
e. ϕ = deviation in azimuth degrees from coincidence of the 135-Hz modulation and the auxiliary reference group
f. γ = bearing to the ground or shipboard station from the point of observation
g. f = pattern rotation frequency in Hz
h. t = time in seconds from the corrected position of the tenth pulse of the X mode main reference group or 15 microseconds (µs) after the sixth pulse of the Y mode main reference group. It is computed from the average position of the twelfth pulse of each auxiliary reference group.

Tolerances

f = 15 Hz ±0.2 percent	A = 0.21 $^{+0.09}_{-0.10}$
θ = 6.0 degree maximum	B = 0.21 $^{+0.12}_{-0.14}$
ϕ = 1.5 degree maximum	Sum of A and B not greater than 0.55

Modulation index (15 Hz and 135 Hz) shall be established at the factory; field adjustment of the two parameters shall not be possible.

3.7.1.3.3 Harmonics. The root sum squared (rss) harmonic content of the 15 Hz component shall not exceed 25 percent, taking all terms through the eleventh harmonic, but excluding the ninth harmonic. The rss harmonic content of the 135 Hz component shall not exceed 30 percent, taking into account the second, third, and fourth harmonic terms.

3.7.1.3.4 Vertical coverage. The maximum carrier energy shall be radiated above the plane perpendicular to the vertical axis of the antenna. The carrier slope between 2 degrees below the horizon and 2 degrees above the horizon will be no more than 0.06 volt per degree normalized to the horizon. Modulation of the 15 Hz and 135 Hz components shall be within the limits specified in 3.7.1.3.2 from the horizon through 30 degrees. Relative energy radiated below the horizon shall not exceed 40 percent of the total. In the region of 0 degrees to -20 degrees, the modulation shall not be less than 7 percent and shall not undergo any phase reversals as referenced to the phase at the horizon. In the region 0 degrees to -45 degrees, the maximum lobe shall be at least 9 decibels (dB) below the peak of the main beam.

3.7.1.3.5 Gain. The peak gain shall be at least 3 dB and not more than 6.5 dB over an isotropic radiator and shall occur with an angle of +35 degrees (referenced to the horizon). Horizon directive gain shall not be less than -2.0 dB over an isotropic radiator.

3.7.1.3.6 Voltage standing wave radio (VSWR). The VSWR of the antenna as measured at the end of 1.524 meters (m) (5 feet (ft)) of M17/86-00001 cable shall not exceed 2.2:1 within the band 962 MHz to 1213 MHz. The M17/86-00001 shall conform to MIL-C-17/86.

3.7.1.3.7 RF power capacity. The antenna shall be capable of operating with a peak power of 60 watts (W) to 3 kilowatts (kW).

3.7.1.3.8 Polarization error. Bearing errors due to horizontally polarized signals shall not exceed 2.0 degrees for the 135 Hz component for 5.0 degrees for the 15 Hz component. Measurements made to determine the extent of this error shall be made with a dipole oriented ±45 degrees with respect to the vertical where reference bearing is established with the dipole in the vertical orientation. Readings shall be taken from the horizon through +10 degrees elevation. Readings taken at -10 degrees elevation shall not exceed 3.5 degrees for 135-Hz component. For the shore configuration, the bearing errors due to horizontally polarized signals shall not exceed 7.5 degrees for the 15-Hz components at -10 degrees elevation.

3.7.1.3.9 Impedance. The impedance of the antenna over the entire transmission and reception band (962 MHz to 1213 MHz) shall be nominally 50 ohms.

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3.7.1.3.10 Azimuth accuracy. The peak azimuth error attributable to the antenna group (excluding the discharge protection rod) shall not exceed 8 degrees for the 15-Hz signal and shall not exceed 1.5 degrees for the 135-Hz signal; both of these limits apply at all azimuths, 0 degree to +360 degrees, and from elevation angles of +2 degrees through +10 degrees.

3.7.2 Control unit. The control unit shall contain control and switching circuitry to enable operation of the antenna as well as monitoring circuitry to continuously survey the operation of certain key functions of the TACAN operation. This unit shall be configured in a single wall mounted box. All controls and indicators necessary for proper operation shall be on the front panel of the control unit. Circuitry associated with the monitoring functions shall be located in the control unit, to the maximum extent possible. Any antenna required to accomplish the monitoring specified herein shall be located in the antenna itself. Any monitoring circuitry which cannot be located in the control unit shall be in the antenna portion. Parameters to be monitored shall be azimuth accuracy, relative power output, azimuth tracking (sense whether or not the antenna is correcting azimuth reference in step with turning of the ship), reference burst presence and internal pulse count, and internal moisture (sense the presense of moisture inside the radome). Relative power output shall be monitored at the 3 kW or the 700 W level depending on the character of the beacon ready signal, (see 3.7.2.2e). Signals required for the monitoring specified herein (except azimuth tracking, azimuth accuracy, and internal moisture) shall be obtained off-air, after radiation by the antenna group. The monitoring function shall provide a two-level failure (see 6.4.1) indication: Fail-soft, indicating equipment malfunction which results in degradation of the TACAN signal, but does not render it unusable or unsafe from a flight safety standpoint, and, Fail-hard, indicating equipment malfunction which renders the TACAN unusable or unsafe (see 3.7.2.1). Status of monitored parameters shall be indicated by a front panel light for each. Instantaneous failures shall be indicated by the front panel lights with a 1-second memory so that positive visual observation may be made. Any indicated failure which remains for 8 seconds or more shall result in initiation of a remote soft or hard failure alarm. This alarm condition and associated panel indications shall be maintained until the reset function is activated. Reset shall result in a complete clearing of all failure indications and repeat of the 8-second time out cycle. An instantaneous digital numeric readout of the electronic ships heading correction plus the magnetic variation setting shall be provided on the front panel. A peak power meter capable of indicating a power normal level in the 60-W to 6-kW range shall be provided.

3.7.2.1 Monitored tolerances. Monitored tolerances shall be as specified in TABLE I.

TABLE I. Monitored tolerances.

Parameter	Fail-soft	Fail-hard
Azimuth accuracy 15 Hz	<10 degrees	>10 degrees
Azimuth accuracy 135 Hz	< 2 degrees	> 2 degrees
Power output	2dB ±1.0dB	
Azimuth tracking		Alarm first through ninth bit,
Main reference group		10 pairs or less for X channels; eleventh pulse or less for Y channels;
Auxiliary reference group		4 pairs or less for X channels; eleventh pulse or less for Y channels
Moisture	90 percent ±5 percent	
Ship heading input	humidity	Loss of ships compass synchro reference signal
Shore criteria		The azimuth alarm circuits shall have an alarm threshold tolerance limit of 0.5 degree to 1.0 degree. The time to activate the alarm shall be 10 seconds ±2 seconds.

3.7.2.2 Remote controls, interlocks, and beacon interfaces. The system of controls and interlocks specified in a through g shall be provided in order to interface the antenna group control unit properly with the TACAN beacon in an operational installation.

a. The MAIN and AUX reference triggers shall be provided to the beacon each by a separate shielded twisted pair. The load is specified in 3.7.3. In addition, both triggers shall be available from BNC panel jacks on the bottom of the control unit.

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b. The SOFT FAIL ALARM signal shall be provided to the beacon by a contact open on a shielded twisted pair. Maximum load to be handled on this line is 28 volts direct current (VDC) 100 milliamperes (mA). A remote alarm inhibit switch shall be mounted to control both the soft and hard alarms. A red indicator light shall indicate when the switch is in the inhibit position.

c. The HARD FAIL ALARM signal shall be provided to the beacon by a contact open on a shielded twisted pair. Maximum load to be handled on this line is 28 VDC, 100 mA.

d. Remote turn-on of the antenna group shall be provided by the application of a remote +28 VDC level signal. The signal and its return line shall consist of a shielded twisted pair. The maximum load to be handled on the return line is 200 mA at +28 VDC.

e. A beacon ready signal shall be provided to the control unit on two shielded twisted pairs where logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum and logic 1 is +2.5 VDC to +5.0 VDC with 1 mA source current at +3 VDC. The logic 00 of this signal will result in lockout of the peak power and reference burst alarms. Logic 11 represents beacon power level of 150 W and will result in lockout of peak power alarm. A 01 signal will represent a 3 kW power and a 10 signal will represent the 700 W power. Maximum load to be handled on these lines is 200 mA at 5 VDC.

f. Antenna band switching and X-Y Mode of operation signals shall be provided to the antenna group on three separate shielded twisted pairs where logic 0 is -0.5 VDC to +0.5 VDC, 20 mA sinking current maximum. Logic 1 is +2.5 VDC to +5.0 VDC with 1 mA source current at +3 VDC. The switching logic shall be as specified in TABLE II.

TABLE II. Switching logic.

Frequency (MHz)	X-Y Mode	Band code	Band
962-993	0	00	A
994-1024	0	01	A
1025-1055	1	00	A
1056-1087	1	01	B
1088-1119	1	10	B
1120-1150	1	11	B
1151-1181	0	10	C
1182-1213	0	11	C

This switching logic encompasses a 0 code for beacon operation in the X Mode and 1 code for the Y Mode. For the shore configuration, the frequency selection shall be performed manually with the frequency bands specified in TABLE III.

TABLE III. Frequency bands.

Band	Frequency (MHz)
A	962 - 1055
B	1056 - 1130
C	1131 - 1213

g. A CONTROLS NORMAL/ABNORMAL signal from the beacon shall be returned by the control unit on one line with a contact closure when the LOCAL/REMOTE switch is in the REMOTE position, and on a second line when the LOCAL/REMOTE switch is in the LOCAL position. Maximum load on this line is 28 VDC, 200 mA.

3.7.2.2.1 Interface cable. The interface cable carrying the signals listed in a through f of 3.7.2.2 shall be Type 25 WU-12 with a maximum length of 15.24 m (50 ft). The mating connector to be supplied shall be a Bendix 10-214628-21P, or equivalent. The controls NORMAL/ABNORMAL signal of 3.5.2.2 shall be available at a separate panel jack at the bottom of the control unit. The signals shall be wired as shown on NAVSEA DWG 503486.

3.7.2.2.2 Gyro input filter. A five-section T filter shall be provided at the synchro input to the control indicator. The filter shall be Capcon Inc., Part No. LMT-115-025-T, or equivalent.

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3.7.3 Azimuth trigger generation. This equipment shall contain circuitry to generate azimuth triggers for use by a TACAN beacon in generation of the north and auxiliary bursts. The triggers from this group shall be synchronous with the modulation produced by the antenna. One fundamental and 8 ninth-harmonic triggers shall be produced for each 360 degree rotation of the antenna pattern. The ninth harmonic trigger pulses shall be spaced 40 degrees ±0.1 degree in azimuth apart, except that the eighth and next succeeding first pulse shall be 80 degrees ±0.2 degrees apart. The fundamental trigger pulse shall occur 40 degrees ±0.1 degree in azimuth after the last (eighth) ninth-harmonic pulse. Trigger pulse amplitude shall be +7.0 V ±1.5 V and -7.0 V ±1.5 V with the shape shown in FIGURE 2, measured across 10 kilohms shunted by 0.005 microfarad. For X channels, synchronization between the fundamental trigger pulse and the modulation from the antenna shall be such that for a bearing of 0 degree, with the ship's heading and magnetic variation inputs at 0 degree, and the receiver south of the antenna, the tenth pulse of the main reference burst (generated by a beacon triggered by the fundamental pulse in question) coincides with the positive slope of inflection of the 15-Hz modulation. The ninth harmonic trigger pulse shall be synchronized with the antenna modulation such that the twelfth pulse of each auxiliary burst (generated by a beacon triggered by the ninth harmonic trigger under discussion) coincides with each positive slope point of inflection of the 135-Hz modulation. For Y channels, the coincidence points shall be midway between the sixth and seventh pulses of the 15-Hz reference group, and shall be at the twelfth pulse of the 135-Hz reference group. The tolerances on each of these synchronization requirements shall ensure azimuthal errors of no more than ±1.0 degree for the coincidence of the 15-Hz modulation and the fundamental trigger and ±0.3 degree for the coincidence of the 135-Hz modulation and the ninth harmonic trigger. Delay compensation may be required in the X channel trigger circuitry to achieve these tolerances. The fundamental trigger shall be produced such that the main reference group occurs when the maximum of the composite radiated signal is directed east of the transponder site. The antenna group shall incorporate circuitry as required to maintain proper azimuth orientation as the ship maneuvers. Ship synchro inputs of one speed shall be utilized for reference, but not as power supplies. Provision shall be made for including magnetic variation in the correction. An internal electrical 0 signal shall be available for alignment or maintenance. The signal shall be controlled by a two position rotary switch which would normally be set in the Ship's Gyro position. Manual control of magnetic variation shall be provided on the front panel for 0 degree to 90 degrees easterly and 0 degree to 90 degrees westerly with deviation in 1-degree increments. An internal control shall provide for system alignment and additional correction to achieve a total of ±180 degrees. The manual control shall display 0 degree to 180 degrees easterly and 0 degree to 180 degrees westerly with deviation in 1-degree increments. The azimuth orientation circuitry shall be all electronic, containing no motors, gears, pick-up arms, and so forth. The corrective azimuth orientation circuitry shall follow a heading change of up to ±6 degrees sinusoidally in a 6-second period with a lag of no more than 0.4. The antenna system shall conform to all requirements of this paragraph at all azimuths (0 degree to 360 degrees).

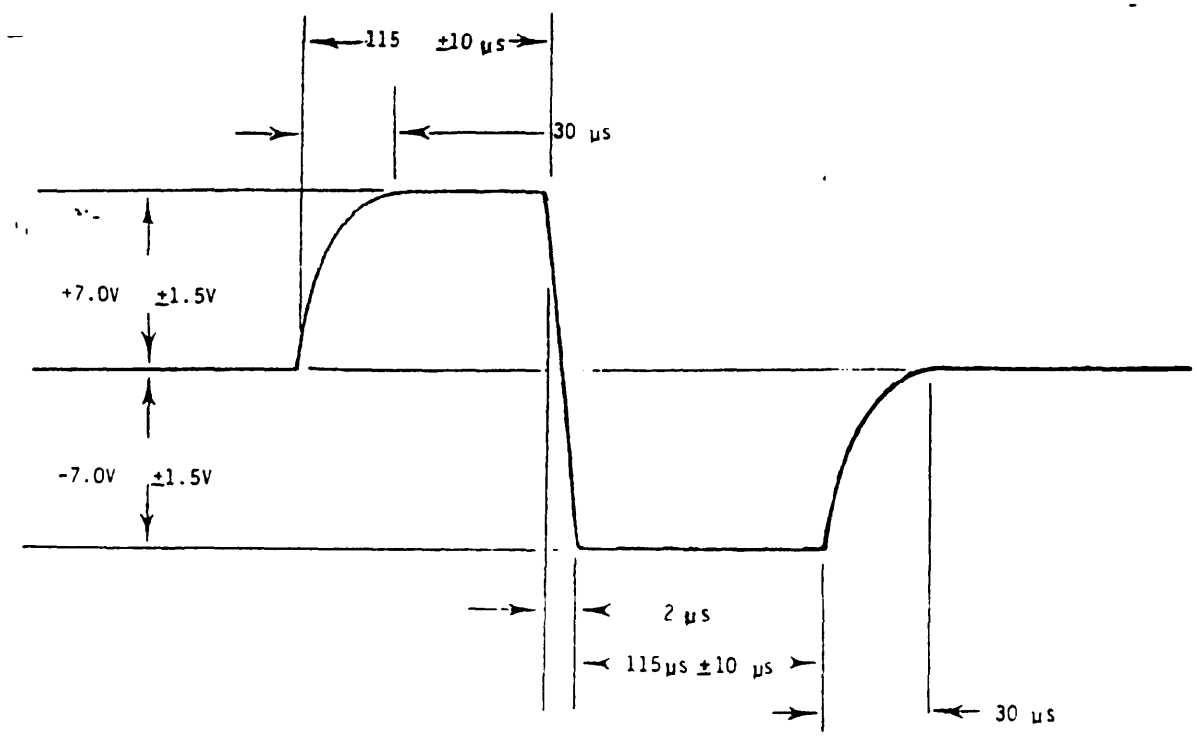


FIGURE 2. Trigger pulse shape.

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3.8 Electromagnetic compatibility (EMC) requirements. EMC shall be as specified in 3.8.1 and 3.8.2.

3.8.1 Interference characteristics. The equipment shall conform to MIL-STD-461, Class A4 requirements, modified as specified in a and b:

a. REO2 - The equipment shall be tested to the broadband radiation emission requirements only and shall not exceed the specification limits in the range:
14 kilohertz (kHz) through 260 kHz by 6 dB
0.5 MHz through 2.4 MHz by 10 dB
2.4 MHz through 25 MHz by 4 dB
b. CEO3 and CEO4 - The equipment shall be tested to the conducted emission requirements, methods CEO3 and CEO4 and shall not exceed the specification limits in the range 15 MHz through 50 MHz by 10 dB. MIL-HDBK-241 may be used for guidance in electromagnetic interference (EMI) reduction in power supply design.

3.8.2 Electromagnetic damage. The equipment shall operate without damage or performance degradation in the normal shipboard electromagnetic environment specified in MIL-HDBK-235, Part I and Part II. This requirement applies to both the mast-mounted portion and the below-deck portion.

3.9 Human engineering. The design shall incorporate the human engineering aspects of MIL-STD-1472 throughout.

3.10 Ventilation and cooling. Ventilation and cooling shall conform to the requirements of the Thermal design paragraph of MIL-E-16400. Blowers or other active mechanical air circulation devices shall not be used.

3.11 Safety. Safety shall be in accordance with the Safety criteria paragraph of MIL-E-16400.

3.11.1 Leakage current. The leakage current of the equipment shall not exceed 5 mA.

3.12 Dimensions and weight. The mast-mounted unit of the antenna group shall not exceed 1371.6 mm (54 in.) in diameter (including radome latches), 711.2 mm (28 in.) in overall height (including mounting plate and adapter) and 32.6 kilograms (115 pounds) in weight (including lightning rod weight). The control unit shall be configured for bulkhead mounting and shall not exceed 546.1 mm (21.5 in.) in width, 406.4 mm (16 in.) in total depth, and 787.4 mm (31 in.) in length excluding the bulkhead mounting flanges.

3.13 Identification and marking. All panel marking shall be engraved. Antenna pedestal marking may be metal stamped.

3.14 Quantitative reliability requirements. The upper test mean-time-between failures (MTBF) (O_Q as defined by MIL-STD-781), of the antenna group shall be 7500 hours. The lower MTBF (O_1 as defined by MIL-STD-781) for the antenna group shall be 2500 hours.

3.15 Maintainability. The antenna group shall have an equipment repair time (ERT), median not exceeding 25 minutes when repair is accomplished at the organizational level of maintenance by replacement of faulty modules (including printed circuit boards (PCBs) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective-maintenance-time includes localization, isolation, disassembly, interchange, reassembly, and checkout for all maintenance tasks. The specified median corrective maintenance time is based on direct access to the mast-mounted unit of the group. The ERT for piece-part repair of repairable modules at the intermediate or depot level of maintenance shall not exceed 120 minutes. The organizational level maintainability requirements shall be based on the capabilities (skill level and experience) of a Navy Electronics Technician, third class (ET-3) as specified in NAVPERS 18068.

3.15.1 Maintenance concept. The maintenance concept for this equipment is for repair by replacement at the module level utilizing the built-in diagnositc and fault location circuits of the control unit, module, chassis test points, and general purpose test equipment.

3.16 Workmanship. Workmanship shall be as specified in 3.16.1 through 3.16.2.2.3.

3.16.1 General workmanship. Workmanship shall be as specified in MIL-STD-454, Requirement 9.

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3.16.2 Screening. Screening shall be as specified in 3.16.2.1 through 3.16.2.3.

3.16.2.1 Pin diodes. Each pin diode used in the AS-3240/URN antenna shall withstand 168 hours at 125°C as a reverse bias screen.

3.16.2.2 Workmanship screen. Workmanship screen shall be as specified in 3.16.2.2.1 through 3.16.2.2.3.

3.16.2.2.1 Vibration screen. The AS-3240/URN antenna and the C-10328/URN control-indicator shall be subjected to a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ from 80 Hz to 350 Hz as specified in 4.8.2.2.1. The rate shall be one antenna group from each month's production.

3.16.2.2.2 Temperature cycling, AS-3240/URN antenna. Each antenna (less lightning arrestor and radome) shall withstand temperature cycling as specified in 4.8.2.2.2.

3.16.2.2.3 Temperature cycling, C-10328/URN control-indicator. Each control-indicator shall withstand temperature cycling as specified in 4.8.2.2.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Safety verification (see 4.5.3)
d. Inspection of preparation for delivery (see 4.9).

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4.3 First article inspection. Unless otherwise specified (see 6.2), four units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV and TABLE V. After Government first article approval, first article units shall be refurbished and pass the production inspection.

TABLE IV. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Weight	3.12	4.5	X			
Dimensions	3.12	4.5	X			
Parts,Materials, and processes	3.2	4.5	X			
Finish	3.2.2.1	4.5	X			
Identification and marking	3.13	4.5	X			
Safety	3.11	4.5.3	X			
General workmanship	3.16.1	4.8.1	X	X		
Performance	TABLE II		X	X		
Temperature sequence						
Low temperature	3.3.1	4.5	X			X
High temperature	3.3.1	4.5	X			X
Humidity	3.3.6	4.5	X			X
Salt fog (spray)	3.3.2	4.5	X			X
Shock	3.3.7	4.5	X			
Vibration	3.3.8	4.5	X			
Enclosure	3.6.5	4.5	X		X	
Steady-state voltage and frequency	3.4.1.1	4.5	X		X	
Transient voltage	3.4.1.1	4.5	X		X	
Transient frequency	3.4.1.1	4.5	X		X	
Spike voltage	3.4.1.1	4.5	X		X	
Power interruption	3.4.1.1	4.5	X		X	
Power and power factor	3.4.1.1	4.5	X			
Leakage current	3.11.1	4.5.4	X			X
EMC	3.8	4.5.1	X			X
Reliability	3.14	4.6	X			
Harmonic current	3.4.1.2	4.5	X		X	
Nuclear air blast	3.3.3[1/]		X			
Thermal survey		4.5.2[1/]	X			
Workmanship screen	3.16.2.2	4.8.2.2		X		
Maintainability	3.15[1/]	4.7	X			

[1/] Required on original development contract only.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.3.3.

4.4.1 Production inspection (Group A). Production inspection shall be performed on each antenna group offered for delivery. Production inspection shall comprise such examination and testing that will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the requirements of TABLE IV, Group A, and the performance tests of TABLE V, except that all RF radiation tests shall be performed at the horizon only. The RF power capacity tests may be demonstrated using a 60-Hz voltage equivalent to the 3 kW peak.

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TABLE V. Performance tests.

Tests	Requirement paragraph
Composite 15-Hz and 135-Hz variable bearing signal	3.7.1.3.2, 3.7.1.3.4, 3.7.1.3.5
Harmonics	3.7.1.3.3
VSWR	3.7.1.3.6
RF power capacity (high potential)	3.7.1.3.7
Polarization error	3.7.1.3.8
Azimuth accuracy	3.7.1.3.10
Monitored tolerances	3.7.2.1
Remote controls, interlocks, and beacon interfaces	3.7.2.2
Azimuth trigger generation	3.7.3
Lightning protection (peak azimuth errors)	3.3.4

4.4.2 Production control inspection (Group B). Production control inspection shall be performed on a randomly selected sample from each month's production. Production control inspection shall include the requirements of TABLE IV, Group B. Production control inspection shall be performed on antenna groups that have passed production inspection. The sample shall satisfactorily complete the production control inspection prior to the release of the month's production for shipment.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be as specified in TABLE IV, Group C, and the RF radiation tests specified in TABLE VI, and shall be measured above and below the horizon. Environmental inspection and RF radiation tests shall be performed on antenna groups that have been subjected to, and have passed, production inspection.

TABLE VI. RF radiation tests.

Tests	Requirement paragraph	ITTAV DWG 1098792 paragraph	ITTAV DWG 1098793 Data sheet
Composite 15-Hz and 135-Hz variable bearing signal	3.7.1.3.2	3.1	4
Vertical coverage	3.7.1.3.4	3.1	3
Gain	3.7.1.3.5	3.1	3
Harmonics	3.7.1.3.3	3.1	5, 6
Polarization error	3.7.1.3.8	3.2	8
Azimuth accuracy	3.7.1.3.10	3.2	7

4.4.3.1 Sampling for environmental inspection. One antenna group shall be selected from each successive 25 antenna groups or fraction thereof, produced. The first sample antenna group shall be selected from the first month's production. The sample shall satisfactorily complete the environmental inspection specified in 4.4.3 prior to the release of the second month's production for shipment.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR), the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed, environmental inspection may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

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4.5 Test methods. The examinations and tests specified in TABLE IV shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.5.1 through 4.5.4.

4.5.1 EMC test. The requirements of 3.8.1 shall be verified by tests performed in accordance with MIL-STD-462.

4.5.2 Thermal survey. The contractor shall conduct a complete thermal survey of the antenna group during temperature environmental testing to determine the existence of hot spots (see 6.4.5) which may prove to be injurious to the useful life of the antenna groups. The survey shall be conducted with the antenna group packaged and enclosed in the proposed final production configuration. The survey shall include a thermal profile of the internal configuration of the antenna group and all critical items at both 25°C ambient and the maximum specified ambient operating temperature. The survey shall be completed and all required corrective measures taken by the contractor prior to approval by the procuring activity. The thermal survey shall be repeated whenever antenna group changes occur which may affect this characteristic.

4.5.3 Safety verification. Examinations, analyses, and tests shall be performed to verify conformance to 3.11.

4.5.4 Leakage current test. The leakage current test shall be performed as specified in 4.5.4.1 and 4.5.4.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE ANTENNA GROUP DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.4.1 Antenna group connections. Each antenna group directly connected to an external power source and units deriving power from the antenna group shall be placed on an insulated surface. All safety ground conductors between the antenna group and units deriving power from the antenna group shall be intact. The safety ground conductor between the antenna group and the source power shall be opened during the test. The antenna group shall be connected as shown in FIGURE 3 if it is connected to single-phase source power.

4.5.4.2 Measurement. Leakage current shall be measured on the antenna group in its normal operating configuration. Antenna group controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the antenna group is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURE 3. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.6 Reliability qualification test (RQT). An RQT shall be conducted on two first article antenna groups in accordance with the general requirements of MIL-STD-781, Test Plan XXIC. Testing shall proceed to a decision in accordance with the accept or reject criteria of MIL-STD-781, Test Plan XXIC.

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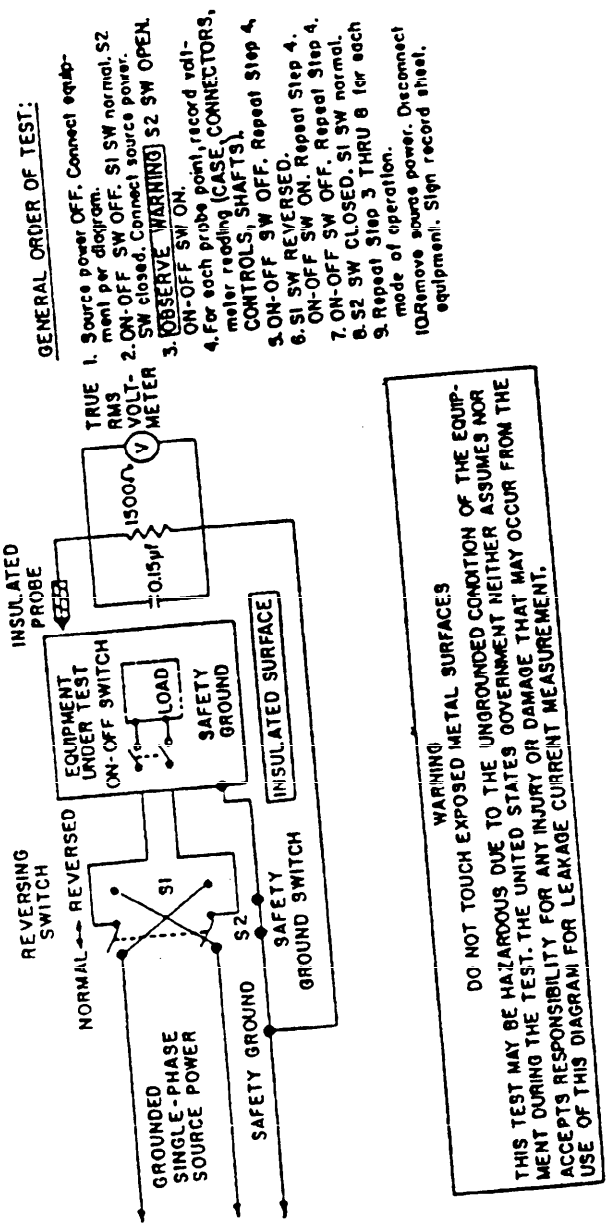


FIGURE 3. Single-phase test diagram for leakage current measurement.

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4.6.1 RQT conditions. The RQT shall be run under the test conditions specified in a and b:

a. Portions of the antenna group located below deck shall be subjected to the environmental conditions of MIL-STD-781, Equipment Category 3A, and the Test conditions matrix table (for shipboard sheltered equipments) of MIL-STD-781.

b. Portions of the antenna group located above deck shall be subjected to the environmental conditions of MIL-STD-781, Equipment Category 3B, and the Test conditions matrix table (for shipboard unsheltered equipments) of MIL-STD-781, except that the operating temperature range shall be -28°C to +95°C.

4.6.2 Rejection. If the test results in a reject decision, the procuring activity shall be immediately notified and have the option, at no additional cost to the Government, to require the contractor to either: a) rerun the entire RQT, or b) rerun portions of the RQT to verify the effectiveness of the corrective action developed by the contractor to remedy causes for the rejection. Successful demonstration of all corrective actions, without any additional failures, shall be a basis for hardware acceptance.

4.7 Maintainability demonstration. Conformance with the equipment and module maintainability requirements shall be demonstrated in accordance with MIL-STD-471, Test Method 4. Only MIL-STD-471, Phase II shall apply.

4.7.1 Equipment maintainability. The equipment maintainability demonstration shall consist of 20 corrective maintenance tasks requiring replacement of modules (or PCBs) or chassis-mounted parts, accomplished at the organizational level of maintenance. Fifty faults shall be determined in accordance with MIL-STD-471, APPENDIX A. The procuring activity, or its representative, shall use the faults as a guide to determine a sample of 20 faults for the maintainability demonstration. Fault simulation for modules shall be accomplished by introduction to faulty parts corresponding to the number selected or by simulating such faults. As part of the demonstration, the accessibility shall be observed to determine conformance with 3.5.3.

4.7.2 Module maintainability. The module maintainability demonstration shall consist of 20 corrective maintenance tasks requiring repair of modules (or PCBs) by replacement of parts, accomplished at the intermediate or depot level of maintenance, as appropriate. A minimum of five faults shall be determined for each repairable module. Fault selection and simulation for each repairable module shall be in accordance with the procedure for the equipment maintainability demonstration. A total of 20 faults shall be determined by the preparing activity for the module maintainability demonstration.

4.7.3 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the equipment maintainability demonstration.

4.7.4 Technical documentation. Technical documentation to be used in the demonstration shall be limited to the technical manual.

4.7.5 Rejection. Failure to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction to all deficiencies, and

b. Upon approval of an approach to the redesign or corrections, the contractor shall implement the approach and repeat the demonstration until an accept decision is reached.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 through 4.8.2.2.3.

4.8.1 General workmanship. The antenna group shall be examined to determine conformance to the requirements of 3.16.1.

4.8.2 Screening. Screening shall be as specified in 4.8.2.1 through 4.8.2.2.3.

4.8.2.1 Pin diodes. Prior to assembly of the antenna parasitic elements, each pin diode used therein shall be subjected to reverse bias operation, at 80 percent voltage rating, for 168 hours at 125°C. At the end of 168 hours at 125°C, the temperature shall be reduced to 25°C over a period of time not less than 1 hour while maintaining 80 percent voltage rating. The rejection criteria shall be as specified in a and b:

a. Any unit exceeding the manufacturer's specified limits

b. ΔI_R ±100 percent of initial value of 1 microampere, whichever is greater

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4.8.2.2 Workmanship screen. Prior to conducting the temperature cycling, the AS-3240/URN antenna and the C-10328/URN control-indicator, shall be subjected to a vibration screen. The top and bottom half of the radome and the lightning arrester may be removed for the test so that components will be visible.

4.8.2.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random, or complex waveform vibration, for an accumulated time of 10 minutes. The AS-3240/URN antenna shall be vibrated in the horizontal axis perpendicular to the surface of the PWBs. All test items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 4. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The test items shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. (Antenna and control interconnected with simulated AZ ref-input signal. RF input not required.) All malfunctions occurring during test shall be localized and corrected and the test resumed. Malfunctions will be determined by the fault indicator lights which may blink during the test but not remain steady on for more than 4 seconds. All other equipments produced that month shall be examined as necessary for the presence of like defects and corrected as necessary.

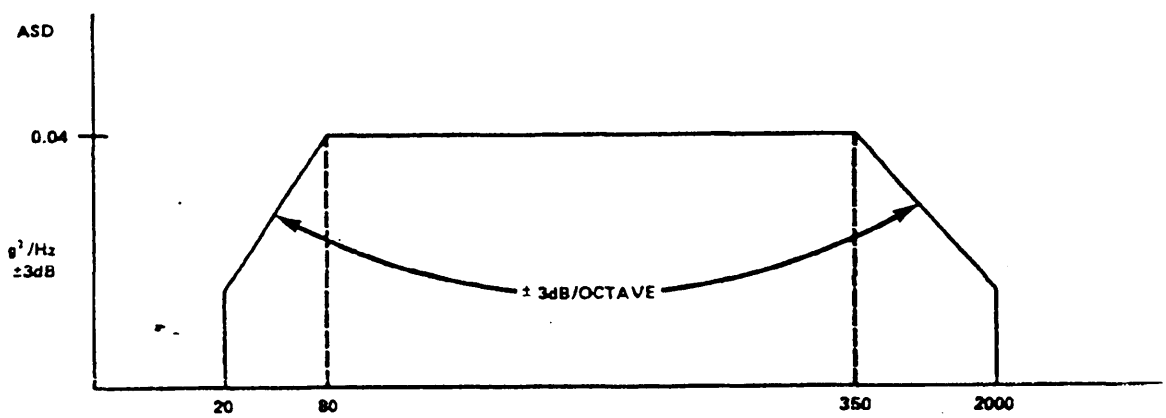


FIGURE 4. Random vibration curve.

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4.8.2.2.2 Temperature cycling, AS-3240/URN antenna. Each antenna shall be subjected to 10 cycles of the temperature curve shown in FIGURE 5. The temperature rate of change shall be not less than 5°C per minute. Antenna power shall be turned on and off at the indicated times. When practical, the antenna doors and radome shall be opened or removed for maximum exposure to the changing temperature. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition. When failures occur, the antenna shall be reworked and the cycling continued for a cumulative total of 10 cycles.

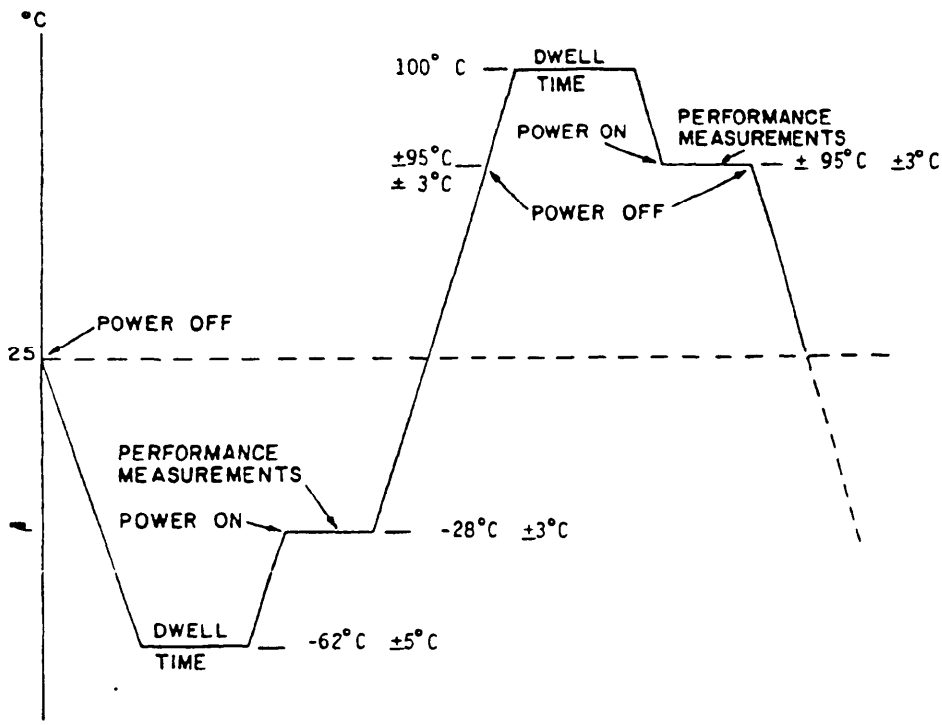


FIGURE 5. One cycle of temperature curve (not to scale).

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4.8.2.2.3 Temperature cycling, C-10328/URN control-indicator. Each C-10328/URN control-indicator shall be subjected to 10 cycles of the temperature curve shown in FIGURE 6. The temperature rate of change shall be not less than 5°C per minute. The control-indicator power shall be turned on and off at the indicated times. When practical, the cabinet panel shall be opened for maximum exposure to the changing temperature. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition. When failures occur, the control-indicator shall be reworked and the cycling continued for a cumulative total of 10 cycles.

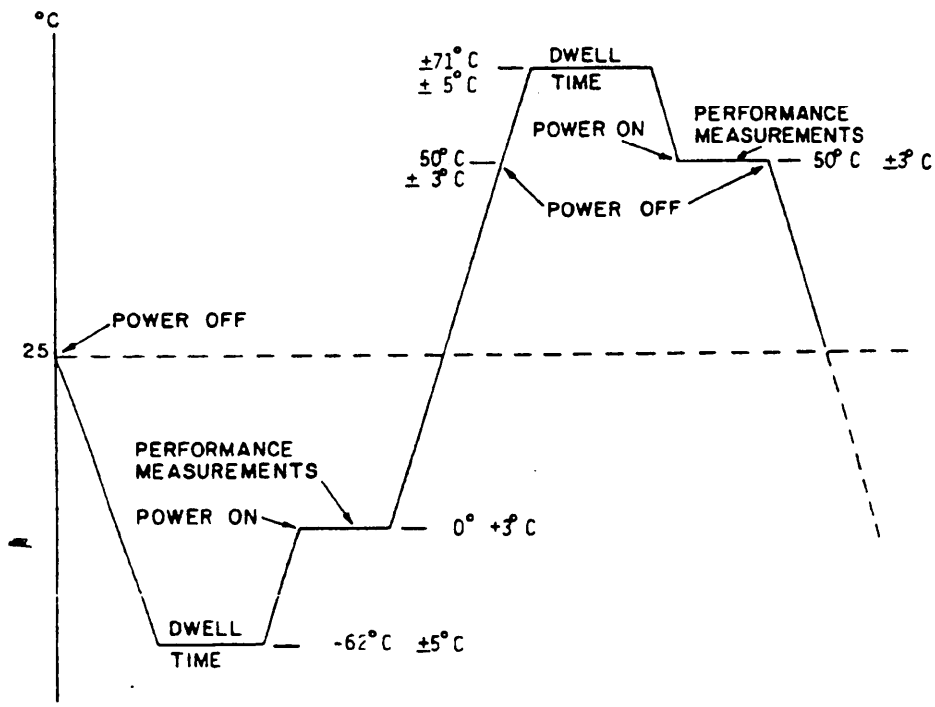


FIGURE 6. One cycle of temperature curve (not to scale).

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4.9 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 or MIL-STD-794, as applicable (see 6.2).

6. NOTES

6.1 Intended use. The antenna group, covered by this specification, is to be interfaced with a family of TACAN beacons installed on board several classes of ships and provide distance, azimuth, and identification information to aircraft.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. When reinspected environmental sample units may not be accepted (see 4.4.3.3)
d. Levels of preservation, packaging, packing, and marking (see 5.1).

6.3 First article. When a first article inspection is required, the items should be first article samples. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 through 6.4.6.

6.4.1 Failure. The term failure refers, in reliability demonstration, to any malfunction or parameter deviation that prevents the equipment from performing to full capacity in all modes, in the specified operational environment, and to the operating parameter levels set forth herein.

6.4.2 Subassembly. Two or more parts which form a portion of an assembly, but having parts which are individually replaceable (such as, modules and PCBs with components).

6.4.3 Assembly. A number of parts or subassemblies or any combination thereof, joined together to perform a specific function and capable of disassembly (such as, audio oscillator, amplifier, synthesizer, crystal oscillator).

6.4.4 Unit. An assembly or any combination of parts, subassemblies, and assemblies mounted together, normally capable of independent operation in a variety of situations. For the purpose of this specification, each base station is considered as a unit.

6.4.5 Hot spot. A hot spot is defined as any item (part or assembly of parts) used in the antenna group including, but not limited to semiconductor devices, PCBs, transformers, resistors, and so forth which attain a temperature in excess of that recommended by the item manufacturer or applicable Military specification, or an assembly of parts that attain a temperature which could cause accelerated degradation of the antenna group.

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4.4.1 Intergranular attack and pitting. From each lot of springs which

6.4.6 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.5 Subject term (key word) listing.

Accessibility
Antenna Group, TACAN, OE-273(V)/URN
Azimuth accuracy
Azimuth trigger generation
Beacon
Bearing signal, variable, composite
15 Hz and 135 Hz
Cable entrances
Cable interface
Control illumination
Control locking switches
Controls, indicators and panel layouts
Distribution statement A
Electrical performance
Electromagnetic capability (EMC)
Enclosure
Environmental requirements
Frequency coverage
Fuse locations
Gain
General purpose electronic test equipment (GPETE)
Gyro input filter
Harmonic current limitation
Harmonics
Human engineering
Humidity
Ice loading and heavy rain
Impedance
Intended use: Shipboard. Provide distance, azimuth, and identification information to aircraft
Leakage current
Lightning protection
Maintainability
Mast-mounting
Microelectronic devices
Moisture protection
Navigational aid
Nonstandard parts
Nuclear air blast
OE-273(V)/URN
Painting
Parts, materials and processes
Pin diodes
Polarization
Power variations
RF power capacity
Reliability
Remote controls, interlocks, and beacon interfaces
Safety
Salt fog (spray)
Shock
Soldering
Temperature cycling
Vertical coverage
Vibration
Vibration screen
Voltage standing wave ratio (VSWR)
Workmanship screen

6.6 Supersession data. This specification includes the requirements of ELEX-A-144A, dated 17 November 1975.

Preparing activity:
NAVY - EC

(Project No. 5985-N543(EC))

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